

Prerequisite MLA02

**Course
Objectives**

The course on reinforcement learning aims to provide the students with the following:

1. Learn how to define RL tasks and the core principals behind the RL, including policies, value functions, deriving Bellman equations (as assets by the assignments, an exam and quizzes)
2. Implement in code common algorithms following code standards and libraries used in RL
3. Understand and work with tabular methods to solve classical control problems
4. Explore imitation learning tasks and solutions
5. Recognize current advanced techniques and applications in RL

On successful completion of the course, the student will be able to:

1. Formulate and analyze basic components of Reinforcement learning for decision-making in various environments.
2. Analyze the concepts of Temporal Difference learning and applications of Deep Q Networks.
3. Optimize various policy-based Reinforcement learning algorithms applicable for different environments.
4. Understand the principles of the model-based RL approaches.
5. Evaluate the effectiveness, scalability, and limitations of RL algorithms and approaches in addressing real-world challenges and applications.
6. Implement basic reinforcement learning algorithms such as Markov Decision Processes, Monte Carlo methods and Temporal Difference learning.
7. Experiment Q-networks using deep learning frameworks to approximate value functions.
8. Work collaboratively in teams to solve complex reinforcement learning problems, and effectively communicate findings and insights.

**Course
Outcomes**

UNIT I FOUNDATIONS

Introduction and Basics of RL- Defining RL Framework and Markov Decision Process -Policies, - Functions and Bellman Equations -. Exploration vs. Exploitation -Code Standards and Libraries used in RL (Python/Keras/Tensorflow) – **Bandit Algorithms**

UNIT II TABULAR METHODS AND Q-NETWORKS

Planning through the use of Dynamic Programming and Monte Carlo – **Control in Monte Carlo** - Temporal-Difference learning methods (TD(0), SARSA, Q-Learning) - Deep Q-networks (DQN, DDQN, Dueling DQN, Prioritised Experience Replay)

UNIT III POLICY OPTIMIZATION

Introduction to policy-based methods -Vanilla Policy Gradient - REINFORCE algorithm and stochastic policy search - Actor-critic methods (A2C, A3C) - Advanced policy gradient (PPO, TRPO, DDPG)

UNIT IV MODEL BASED RL

Model-based RL approach - Analytic gradient computation - Sampling-based planning - Model-based data generation - Value-equivalence prediction - Model-based policy optimization.

UNIT V RECENT ADVANCES AND APPLICATION

Hierarchical Reinforcement Learning – Hierarchical Abstract Machines – MAXQ - Meta-learning -Multi-Agent Reinforcement Learning - Partially Observable Markov Decision Process - Ethics in RL- Applying RL for real-world problems

TEXT BOOKS

1. Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", Second Edition, MIT Press, (2019).
2. Li, Yuxi. "Deep reinforcement learning." arXiv preprint arXiv:1810.06339 (2018).
3. Wiering, Marco, and Martijn Van Otterlo. "Reinforcement learning." Adaptation, learning, and optimization 12 (2012).

REFERENCES

1. Russell, Stuart J., and Peter Norvig. "Artificial intelligence: a modern approach."Pearson Education Limited, 2016.
2. Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. "Deep learning." MIT press, 2016
3. David Silver's course on Reinforcement Learning ([link](#))

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